

ETHICS AND PROFESSIONALISM HU112

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INTELLECTUAL PROPERTY RIGHTS² AND COMPUTER TECHNOLOGY

WHAT IS INTELLECTUAL PROPERTY (IP)?

- IP broadly describes tangible things such as **ideas, inventions, technologies, artworks, music, and literature** to which one can claim ownership.
- **Intellectual property rights** (IPR) are legal rights bestowed on an individual or a group that created, designed, or invented the activities or processes which led to the intellectual property in domains such as science and technology, business, industry, and the arts.
- These legal rights **protect the moral and economic rights of the creators**, in addition to **the creativity and dissemination of their work**.

COMPUTER PRODUCTS AND SERVICES

- **Computer products** consist of those parts of the computer you can see and touch (e.g., the keyboard, CPU, printer, and monitor).
 - They have **tangible form and intrinsic value**.
- **A service** is an act carried out on behalf of someone, usually a customer.
 - They have **intrinsic value** to the customer or recipient but have **no tangible form**.
 - A computer service can take the form of repairing a computer product and/or configuring and installing a computer network, neither of which has a tangible form but does offer considerable intrinsic value to the owner.

COMPUTER PRODUCTS AND SERVICES (CONT'D)

- Computer software, however, cannot be so easily classified as either a product or a service.
- Since it entered the marketplace, therefore, legal protection of computer software has been very problematic.

COMPUTER PRODUCTS AND SERVICES (CONT'D)

- Computer software is a set of logical instructions to perform a desired task.
- This logical sequence follows an **algorithm**.
- The development of this sequence of instructions goes through the following phases:
 1. **Logic map**: The plan of the idea, process, or an algorithm to accomplish the task. A plan is a **flowchart** of logic control with four major stations, namely, the input, output, processing, and decision box connected by arrows indicating the direction of flow of logic.

COMPUTER PRODUCTS AND SERVICES (CONT'D)

2. *The source code*: Turning the flowchart into a set of instructions using a programming language of choice.
3. *The object code*: The second stage of the implementation of a flowchart in which the source code, with the help of either a compiler or an assembler, is turned into strings of zeros and ones. This step is not yet fully an operational form of the program because at this stage the **program lacks variable addresses and missing library routines.**

COMPUTER PRODUCTS AND SERVICES (CONT'D)

4. *Memory-based executable code*: The object code is passed to the **linker**, another one of the system programs, whose **job is to look for all missing variable addresses and library and personal routines and include them in the object code to produce a load module**. The load module is now ready to be loaded if execution is needed.
5. *Microcode*: This code is not loaded in physical memory. It is **coded and loaded on ROM** (the read-only memory of the computer, which cannot be written on by the user) or burned into the computer hardware at the time of manufacture.

COMPUTER PRODUCTS AND SERVICES (CONT'D)

- Our references to computer programs here, therefore, include **memory-based** and **hardware-based** programs together with the technical manuals and all related documentation.
- Note that computer software, if it is considered at the **execution stage without the technical documentation**, **has an intrinsic value both to the developer and to the buyer**, but it may not have a tangible form unless you consider the medium it is on (e.g., the CD).
- That is, **the manuals and CDs** all have a **tangible form** and **probably some intrinsic value**, but the **program itself does not have a tangible form**, although it has **the most intrinsic value**.
- In this case, we can **classify such software as a service**.

COMPUTER PRODUCTS AND SERVICES (CONT'D)

- Not having a tangible form, however, does not by itself rule out software as a product.
- Courts have defined things such as energy and leases as products, although none has a tangible form.
- So there are cases when we can consider software as a product.

COMPUTER PRODUCTS AND SERVICES (CONT'D)

- Jim Prince puts software into three categories:
 1. **off-the-shelf software** (aka “**canned**” software) such as Windows and others that one can buy ready to use with no alterations allowed by the producer
→ Considered to be a product
 2. software specifically ordered by the customer from the software house or producer to fit the customer's very specific needs
→ Considered to be a service

COMPUTER PRODUCTS AND SERVICES (CONT'D)

3. software the customer buys off the shelf, but with changes allowed, or the customer adds certain parts to the software to meet some specific needs
→ Considered to be a new class "mixed case"

FOUNDATIONS OF INTELLECTUAL PROPERTY

- Gaining the skills to provide computer technology products, services, and software requires a considerable investment in both time and money.
- Thus, the individuals who do this work should gain financial rewards for their efforts.
- Such rewards create an atmosphere of creativity and competitiveness, which in turn creates jobs that drive the economy.
- This creativity must therefore be protected.

FOUNDATIONS OF INTELLECTUAL PROPERTY (CONT'D)

- To do this, a specific set of rights, collectively known as intellectual property rights (حقوق الملكية الفكرية), has been recognized, and laws have protecting intellectual rights been enacted and extended to cover software by different countries and groups of countries to protect those rights.
- Intellectual property rights form a wide scope of mechanisms that include:
 - copyrights (حقوق الطبع والنشر)
 - patents (براءات الاختراع)
 - trademarks (العلامات التجارية)
 - protection of trade secrets (اسرار التجارة)
 - personal identity rights (حقوق الهوية الشخصية)

COPYRIGHTS (حقوق الطبع والنشر)

- Copyright is a right, enforceable by law, accorded to an inventor or creator of an expression (عن طريقة تعبير).
- Such expressions may include creative works:
 - artistic, pictorial, graphic, and sculptural works;
 - musical and sound recordings;
 - audiovisual works including television and motion pictures;
 - literary works and other printed materials such as books, greeting cards, journals, flyers, and leaflets;
 - architectural works;
 - the list goes on.

COPYRIGHTS (CONT'D)

- In general, every original work that has a tangible form and is fixed in a medium is protectable under the copyright law.
- Such works **must be expressions, not ideas.**
 - من أهم مبادئ حقوق الطبع أنها لا تحمي الأفكار وإنما تحمي تعبير المؤلف عن الأفكار.

COPYRIGHTS (CONT'D)

- Works in the Public Domain
 - When the copyright on a work expires, that work goes into the public domain.
 - Other works in the public domain include those owned by governments, non-copyrightable items we listed earlier such as ideas and facts, works intentionally put in the public domain by the owner of the copyright, and works that lost copyrights for various reasons before the copyrights expired.
 - Works in the public domain are not protected by the copyright law and can be used by any member of the public without prior permission from the owner of the work.

COPYRIGHTS (CONT'D)

- Application for a Copyright
 - For authors and creators of works who need this kind of protection, the process begins with an application to the copyright office.
 - Each country's copyright office has different requirements.
 - Copyright Office requires an applicant to include with the application a copy of the work for which a copyright is sought and to file for copyright within 3 months of the first distribution of the work.

COPYRIGHTS (CONT'D)

- Upon receipt of the application by the Copyright Office, it is reviewed to ensure it meets the three criteria of **originality** (الأصالة), **fixation** (التثبيت), and **expression** (التعبير) for the issuing of a copyright.
- **Fixation** (now the most controversial element in the current debate about intellectual property rights in the digital age), refers to **the tangible form in which the creation is perceived by others and defines the domain and parameters of such a creation.**
- For example:
 - computer programs are fixed in binary code;
 - performing arts such as drama, the script is the fixation;
 - fine arts, the painting or the sculpture is the fixation.

COPYRIGHTS (CONT'D)

- **Originality** requirements differentiate among facts, ideas, and expressions as creations.
 - Facts are considered common property for all humanity; no one has a right to an unknown or known fact because it is not considered an invention; the same principle applies to theories, mathematical, scientific, and others.
 - Ideas, as are facts, are also considered common property and therefore are not copyrightable.
- Thus, **originality** is only possible through **expressions**.
- **Such expressions may include ideas, theories, and other inventions.**

COPYRIGHTS (CONT'D)

- Duration of a Copyright
 - In Egypt and the US, and many other countries the work remains protected by copyright laws for the **lifetime of the author plus 50 years**.
 - In the case of more than one author of the work, the protection lasts for **the lifetime of the longest living author plus 50 years**.

PATENTS (براءات اختراع)

- Patents protect inventions or discoveries.
- In many countries, patent protection rights, as are those of copyrights, are provided for by the constitution.
- The laws give inventors or discoverers the exclusive rights to make, use, or sell their invention or discovery for a specific period of time, so long as there was full disclosure of the invention or discovery at the time the patent application was filed.
- Because of the disclosure requirement that every patent applicant must meet, the patent is more of a contract between the inventor or discoverer and the government.

PATENTS (CONT'D)

- For this contract to be binding, each party to the contract must keep its part:
 - the **government's part** to protect the exclusive rights of the inventor or discoverer while such a person recovers his or her investments for a period of time,
 - that of the **inventor or discoverer** to make a full disclosure to the government of the invention or discovery.

PATENTS (CONT'D)

- What is Patentable?

- An invention or discovery is patentable if it meets two basic requirements.
- The **first requirement** is that the invention or discovery for which the patent is sought is **new and useful**, or is a **new and useful improvement of any of the following**:
 - process,
 - manufacture (covering all products that are not machines), and
 - machine (covering all mechanisms and mechanical products and composition of matter, which includes all factory-manufactured life forms).

PATENTS (CONT'D)

- The **second requirement** is that the invention or discovery must satisfy the following four conditions, and all must apply:
 1. **Utility (منفعة)**: An invention or discovery **serves a basic and minimum useful purpose** to the general public or to a large percentage of the public without being a danger to the public, illegal, or immoral.
 2. **Novelty (حداثة)**: The invention or discovery for which a patent is sought **must be new, not used, known, or published somewhere before**.

PATENTS (CONT'D)

3. *Nonobviousness* (اللاوضوح): The invention or discovery for which patent protection is sought **must not have been obvious to anyone with ordinary skills to produce or invent in its disclosed form.**
4. *Disclosure* (كشف/إفشاء): There must be **adequate disclosure of the product for which a patent is sought.** Such a disclosure is often used by the Patent Office in its review to **seek and prove or disprove the claims on the application form and also to enable the public to use the invention or discovery safely and gainfully after the period of protection.**

PATENTS (CONT'D)

- Application for a Patent
 - In many countries, the process of obtaining a patent begins with the filing of an application with the patent office.

PATENTS (CONT'D)

- Duration a Patent

- After the review process is completed, the patent is then issued to the applicant **for the invention, and only for that invention, not including its variations and derivatives.**
- The protection must last for a number of years,
 - 17 years in the United States
 - 20 years in Egypt (starting from the date of application).
- During this time period, the patent law protects the inventor or discoverer from competition from others in the manufacture, use, and sale of the invention or discovery.

PATENTS (CONT'D)

- The patent law does not protect ideas, but only the process of carrying out an idea.
- Competitors may take the same idea and use a different process to arrive at their own inventions or discoveries, which can then be patented as novel.

TRADE SECRETS (سر التجارة)

- A trade secret is information that gives a company or business a competitive advantage over others in the field.
- The generic definition is that it is a collection of information in a given static format with strategic importance.
- The format may be:
 - a design expressing the information (design process),
 - a formula representing the collection of information,
 - a pattern,
 - a symbol, or
 - an insignia representing the information.

TRADE SECRETS (CONT'D)

- Whatever the format the collected information takes, it **must have given or offered an advantage to the owner which places that owner a degree above the competition.**
- In the United States, in contrast to the other intellectual properties we have described so far, trade secrets have no federal protection.
- All trade secret laws are state laws.

TRADE SECRETS (CONT'D)

- Characteristics of Trade Secrets

- A trade secret is characterized by the following:
 1. The **extent to which the information is known outside the business**. If many people outside the company or business know or have access to the collection of information that constitutes the trade secret, then it is no longer a trade secret.
 2. The **extent of measures taken by individuals possessing the trade secret to guard the secrecy of the information**. If the information is to remain known by as few people as possible, **there must be a detailed plan to safeguard that information and prevent it from leaking**.

TRADE SECRETS (CONT'D)

3. The **value of the information to the owner and to the competitor**. If the collection of information forming the trade secret has little or no value to the competitor, then it can no longer be a trade secret because it offers no definite advantage to the owner over the competitor.
4. The **amount of effort or money spent by the owner to develop or gather the information**. Usually the more money the developer puts in a project, the more value is placed on the outcome. Because there are some information or project outcomes that do not require substantial initial investments, the effort is what counts here.

TRADE SECRETS (CONT'D)

5. The ease or difficulty with which the information could be properly acquired or duplicated by others. If it will take much effort and money to duplicate the product or the information, then its value and therefore advantage to the competitor diminishes.
- The conditions that characterize a trade secret are in direct conflict with the requirements of a patent.
 - Remember the main requirement for obtaining a patent is the full disclosure of all the information surrounding the product and its workings; this directly conflicts with the need for secrecy in trade secrets.
 - So, the patent applicant cannot claim a patent and at the same time claim protection using the trade secret laws.

TRADE SECRETS (CONT'D)

- Duration of Trade Secrets
 - Have an indefinite life of protection so long as the secrets are not revealed.

TRADEMARKS (العلامات التجارية)

- A label identifying a product or service.
- It is a mark that attempts to distinguish a service or a product in the minds of the consumers.
- The label may be any word, name, picture, or symbol.
- It is very well known that consumers tend to choose between products through association with the product's brand name.
- Because trademarks are used by consumers to choose among competing products, they are vigorously protected by their owners.

TRADEMARKS (CONT'D)

- Differing from patents and copyrights, however, trademarks are not so protected and preserved in constitutions.
- Although:
 - the patent gives owners the exclusive right to use, sell, and make use of the invention or discovery, and
 - the copyright law gives owners the exclusive rights to copying their works,
 - the trademark gives its owner the right to prevent others, mostly competitors, from using the same or similar symbol to market their products.

TRADEMARKS (CONT'D)

- Categories of Trademarks

- Trademark is a general term that includes a **service mark**, a **certification mark**, and a **collective mark**.
- A **service mark** is usually used in the sale or advertising of a service. It is supposed to uniquely identify that service.
- A **certification mark** is used as a verifier or to authenticate the characteristics of a product, a service, or group of people who offer a certain service. For example, colleges attach seals to diplomas as marks to certify the educational attainment of the holders.
- A **collective mark** is mainly used by a group of people to indicate membership in an organization or association (e.g., PhD or MD to indicate they belong to those groups).

TRADEMARKS (CONT'D)

- Characteristics of Trademarks

- A picture is worth a thousand words.
- So, it is assumed by trademark owners that a symbol is worth 1000 words.
- General wisdom is that the more recognizable is the mark, the more valuable will be the product or service.
- Trademarks as symbols of sales to the consumer are generally supposed to tell the consumer about the services or the products they are intended to boost.

TRADEMARKS (CONT'D)

- Trademarks could be put in one of the following characteristic groups:
 - Arbitrary marks
 - Suggestive marks
 - Descriptive marks
 - General marks

TRADEMARKS (CONT'D)

- Registration of Trademarks

- An application for a trademark must contain and present all relevant information.
- It must also describe the product or service for which the trademark is being sought, the class of goods and services, and the date of first issue of the mark.
- Marks are registered only if they meet certain criteria.
- The core requirement is that **the mark must not cause confusion with similar marks used by others.**

TRADEMARKS (CONT'D)

- A mark is registered as a trademark only if it meets the following criteria:
 1. It must be in **good “taste” for the public**: not immoral, deceptive, or illegal.
 2. It must **not have suggestive connotations (دلالات) to its origin**.
 3. It must **not be a symbol of any recognized country**.
 4. It must **not use people’s likenesses**, either after death or if living, **without prior consent**.

TRADEMARKS (CONT'D)

- Duration of a Trademark

- In Egypt and the US, a valid trademark is protected for **10 years**. If an extension is needed it can be granted for another 10 years.

• المادة 90:

مدة الحماية المترتبة علي تسجيل العلامة عشر سنوات وتمتد لمدة أو مدد مماثلة علي طلب صاحبها في كل مرة خلال السنة الأخيرة من مدة الحماية مقابل سداد الرسم المستحق علي طلب التسجيل لأول مرة.

OWNERSHIP

- An idea is novel if it is original, authentic, and new. Inventiveness, creativity, and discoveries are born of individual ideas.
- Good ideas are the source of substantial benefits to individuals and the public.
- Before an idea can be useful, however, it must be put into utilizable form, either as a process or as an application.
- It is this idea in a utilizable form that is the core of intellectual property rights.

OWNERSHIP (CONT'D)

- In many countries the owner of such an application for the idea has a set of legal rights to its expression as a creation, work, invention, discovery, information, or any other form.
- Via the copyright law, the patent law, the trademark law, and trade secret statutes, governments have indicated that they can protect owners' rights under certain conditions, and, therefore, legal ownership of creations, discoveries, information, inventions, and the like, is protectable.

INTELLECTUAL PROPERTY CRIMES

- An intellectual property crime (IPC) is the act of infringement on the rights of the owners of the intellectual property.
- IPC refers to all activities that involve **infringement (التعدي)**, **counterfeiting (التزوير)**, and **piracy (القرصنة)** of products and services for profit.
- IPC also includes **misappropriation (الاختلاس)**, **misrepresentation (التحريف)**, **cybercrimes (الجرائم الإلكترونية)**, **corruption and bribery (الفساد)**, and **espionage (التجسس)**.
- The cost of intellectual property crimes to industry and nations is huge.

INTELLECTUAL PROPERTY CRIMES (CONT'D)

- Technological advances have allowed these crimes to grow like wildfire in the past decade because committing these crimes is much easier and the field of crimes has become global, thus decreasing the threat of apprehension (التخوف).
- Technology has also increased these crimes because duplicated products are easy to make and the costs are low.

INTELLECTUAL PROPERTY CRIMES (CONT'D)

Infringement

- An infringer (المتعدي أو المنتهك) is someone one moving within the protected domain to claim rights for the use of someone else's manifestation of an idea without permission from the holder of the rights.
- There are three types of infringements:
 1. Direct infringement (التعدي المباشر): The infringer knowingly or otherwise makes, uses, sells, or copies a protected item without any alteration.

INTELLECTUAL PROPERTY CRIMES (CONT'D)

2. **Inducement infringement** (الحث علي التعدي): The infringer intentionally supports infringement activities on a protected item without individually taking part in the infringement activities.
3. **Contributory infringement** (المساهمة في التعدي): The infringer takes part in the infringement of a protected item.

INTELLECTUAL PROPERTY CRIMES (CONT'D)

The First Sale Doctrine (عقيدة)

- A copyright owner under the first sale doctrine has the right to distribute copies of copyrighted materials by means of sale, transfer of ownership, rental, release, or by any other means.
- Anybody else who uses that work without either a lease or license is an infringer.

INTELLECTUAL PROPERTY CRIMES (CONT'D)

The Fair Use Doctrine (عقيدة)

- The fair use doctrine establishes a bridge between the protection of rights of artists, authors, inventors, or discoverers to benefit from their works and the basic rights of the community to gain from each member's contributions for the betterment of all and the upholding of the principle of economic competition.
- The use of copyrighted material is considered fair if it does not exploit the commercial value of the work.

INTELLECTUAL PROPERTY CRIMES (CONT'D)

- There are four ways to judge whether the use of an invention, discovery, or work is fair:
 1. The purpose of use, whether commercial or educational
 2. Nature of use
 3. Percentage of use
 4. The effect of use on the commercial value of the invention, discovery, or works

STRATEGIES OF PROTECTING IP

- The **burden of safeguarding** the intellectual property rights of an individual **is with that very person owning the work.**
- It is the duty of individual owners of copyrights, patents, trade secrets, and trademarks to devise strategies to safeguard these rights.
- Various methods have been used by individuals and companies who hold these rights to defend themselves.
 - Large companies and individuals have been known to use methods ranging **from spying on competitors and suspected infringers using private undercover operatives to collaborating with government officials to check on imports and exports.**

STRATEGIES OF PROTECTING IP (CONT'D)

- Some companies **call in their respective governments when they suspect foreign infringements**. When governments step in, they negotiate joint policing within the respective countries and sign treaties to protect these rights.
- Within the United States, some corporations, especially **software companies and computer chip manufacturers**, have **started using local law enforcement agencies to raid suspected infringers** both in the United States and in other countries.
- **Education campaigns via the mass media** are also being used as a viable and effective protection strategy.

PROTECTING COMPUTER SOFTWARE UNDER THE IP

- Algorithms and ideas are not classified as intellectual property and, therefore, are not protected in any way.
- Ideas and algorithms belong to everybody, and no one can put a claim on them.
- **Software**, although it comes from and strictly follows an algorithm, **is not considered an algorithm but rather a manifestation, an expression of that algorithm.**
- Many people may have many different ways of expressing, and therefore representing, the same algorithm and hence have different programs that are considered clear intellectual property and are, therefore, protectable.

PROTECTING COMPUTER SOFTWARE (CONT'D)

- Computer products, in particular computer software, are more elusive and thus have been presenting many problems for those seeking protection under the intellectual property rights law.
- The difficulty with software protection comes from the difficulty in categorizing it.
- As was said earlier, software can be a product, a service, or a mixture of both.

PROTECTING COMPUTER SOFTWARE (CONT'D)

- The biggest problem concerning computer software and intellectual property rights (IPR) is **software piracy**.
- Software piracy **is the act of copying, distributing, or using proprietary software**.
- This act is and **has been illegal** ever since software began to be protected by law after software manufacturers started filing for patents and copyrights for their products and creations.

PROTECTING COMPUTER SOFTWARE (CONT'D)

Protection of Software Under Copyright Laws

- Computer software , along with its documentation, can be protected under the copyright laws.
- According to Section 101 of the 1980 U.S. Copyright Amendment, a **computer program is defined as** “*a set of statements or instructions to be used directly or indirectly in a computer in order to bring about a certain result*”.
- This statement automatically implies that such a set of instructions or statements is a written creative work by someone, hence a literary work.

PROTECTING COMPUTER SOFTWARE (CONT'D)

- Therefore, copyright laws that protect creative works also protect computer programs, including technical manuals and documentation.
- The developer of a program thus has protective rights to:
 - authorize others of his or her choice to reproduce the format of the program en masse at any time,
 - update and upgrade the program, and
 - distribute it for free, or to sell or transfer ownership.
- The copyright laws in this case protect the source code, the object code, and the executable codes, including the manuals and documentation, from illegal copying and piracy.

PROTECTING COMPUTER SOFTWARE (CONT'D)

- Although the developer has such rights, **there are limitations or exceptions** in some instances.
 - For example, a buyer of such a program has some protected rights also. **The buyer has the right to make another copy as a backup, provided that a copy is not used on another machine.**
 - Software registration for copyright, however, does not stop a software developer's worries because **by registering software, the developer is opening up the secrets of the whole project as part of the requirements of the copyright issuance.**
- So before a **developer goes ahead with a copyright application he or she should weigh the pros and cons of copyright protection.**

PROTECTING COMPUTER SOFTWARE (CONT'D)

Protection of Software Under Patent Laws

- Patents protect the processing of the idea; they also protect the implementation of the idea as a machine, a manufacturing process, or a composition of matter or materials.
- Under these conditions, computer hardware components, by their very nature, are protected under the patent laws.
- Software also may be protected under the patent laws under certain circumstances.
- Under these conditions, how can software be protected?

PROTECTING COMPUTER SOFTWARE (CONT'D)

- This is a difficult question, since some patent issues for computer programs are not yet settled.
- There are various reasons for the debate, among which are the following:
 1. The requirement of the patent system for **total disclosure of all information pertaining to the application of the patent is still a big issue**. Given that the patent protection lasts for 17 years and 2-year-old software is as old as software can get and still be really viable, requiring developers to disclose the secrets of their software before the 17-year deadline opens them up to stiff competition and better variations of the applicant's software format.

PROTECTING COMPUTER SOFTWARE (CONT'D)

2. Most of the computer programs on the market are simple one-person ventures. Such persons, many of them independent-minded individuals, may not support yet let alone be able to afford the expense of the patent process application.
3. It has been and still is very difficult, as we saw earlier in this section, to prove to courts and patent offices that algorithms are processes and therefore a form of manifestation of an idea and not mere mental gymnastics that any human being can do and, therefore, not patentable because mental steps and mathematical formulas are not patentable items.

PROTECTING COMPUTER SOFTWARE (CONT'D)

- Although computer programs are not suited for patent protection, there have been successful applications that have received patents.
- For specific examples of some of these cases, see Gervaise Davis's book *Software Protection*.

PROTECTING COMPUTER SOFTWARE (CONT'D)

Protection of Software Under Trademarks

- The main purpose of a trademark is to make the product stand out against the competitors.
- All hardware companies and a few software concerns such as Microsoft have their trademark protected under the trademark laws and statutes.
- But how is a mark or symbol be used to protect computer programs?

PROTECTING COMPUTER SOFTWARE (CONT'D)

- The protection of computer programs by trademarks is achieved through self-realization by the infringer that it is not easy to copy, change, or redistribute copies of well-known software works.
- So for big-name software developers, this realization by would-be infringers works far better than law enforcement.
- But the trick does not work all the time, especially in countries where this sort of realization does not have as much appeal because of lack of publicity of the products.
- However, there are no effective global trademark laws to protect computer programs.

PROTECTING COMPUTER SOFTWARE (CONT'D)

Protection of Software Under Trade Secrets

- A trade secret is information one has about a manifestation of an idea and that no one should disclose or use for the benefit of themselves or a competitor.
- How do these laws help protect computer products, especially software?

PROTECTING COMPUTER SOFTWARE (CONT'D)

- The manifestation of an idea into a computer program usually starts with the blueprint and flowchart, as we saw earlier.
- This process then goes through the remaining stages of object code and executable code.
- One's knowledge of the process anywhere during these stages forms a trade secret and should not be revealed for personal gain or to a competitor.

PROTECTING COMPUTER SOFTWARE (CONT'D)

- It is generally known to computer programmers and software developers, as it is known to all hardware engineers, that **once the blueprint and flowchart of a computer program are known, it is easy to develop a program.**
- It is, therefore, of the **utmost importance that at the early software development stages, the blueprint and flowchart not be known outside design circles.**

TRANSNATIONAL ISSUES AND INTELLECTUAL PROPERTY

- A number of studies concerning the international IP system show that there is **an extensive and growing number of losses being incurred by businesses in the developed world as a result of non-enforcement of IP laws in the developing world.**
- The developed world is charging that the lack of IP legislation in some development countries and the absence of enforcement in others are amounting to sanctioning pirates and **leading to losses amounting to tens of billions of dollars' worth of goods of multinational corporations every year.**



*THINK DIFFERENT
THINK PROFESSIONAL
THINK ETHICAL*